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EECS 1011: Computational Thinking Through Mechatronics

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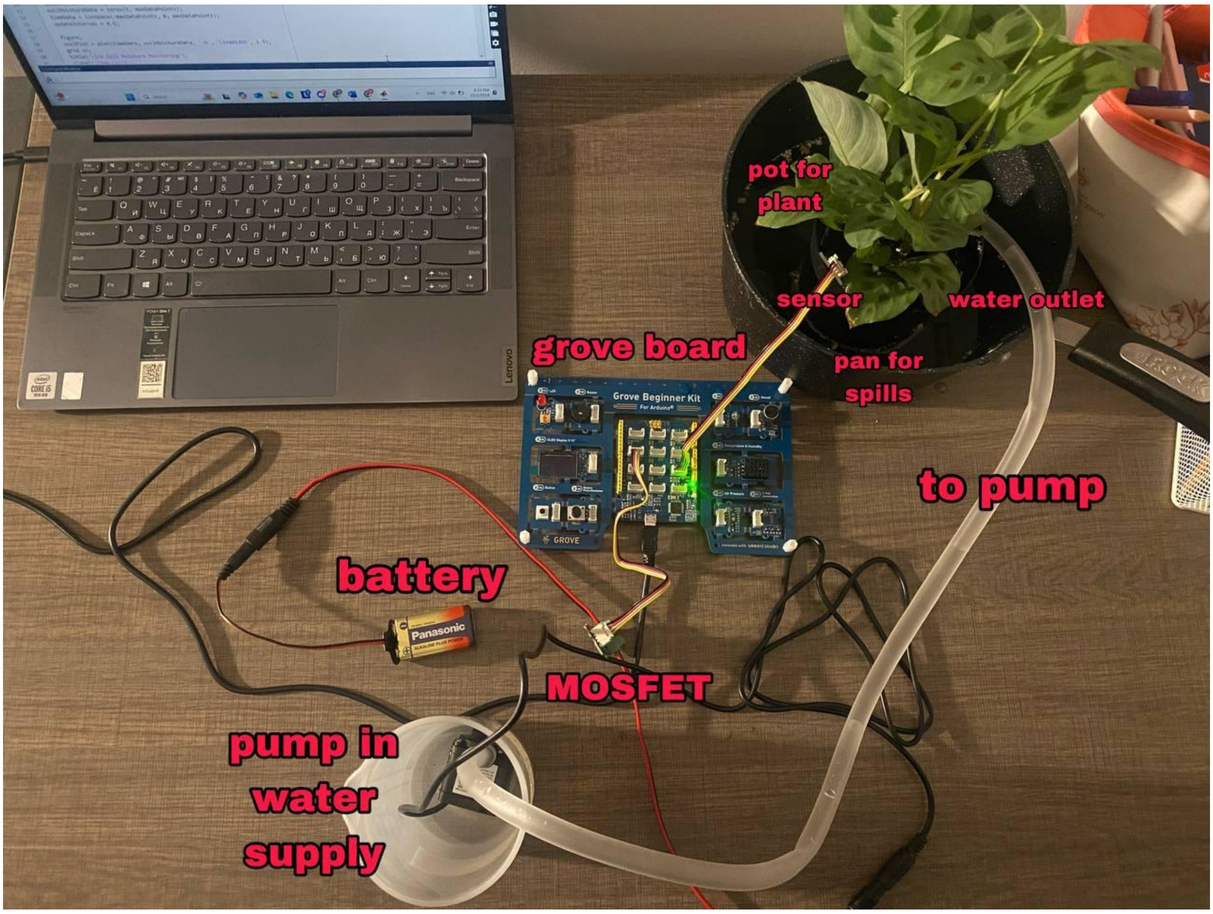
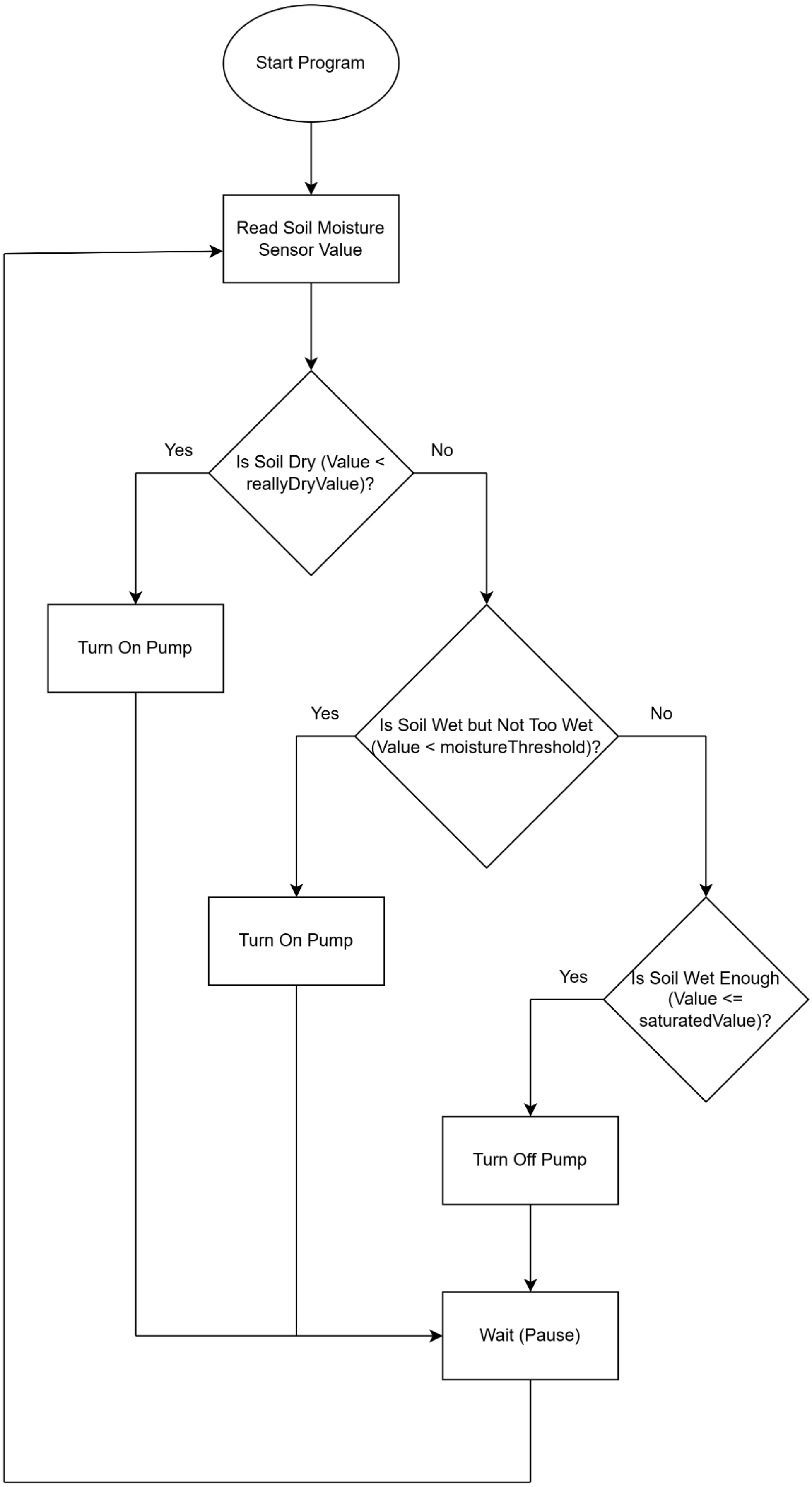
Main Project

Abstract

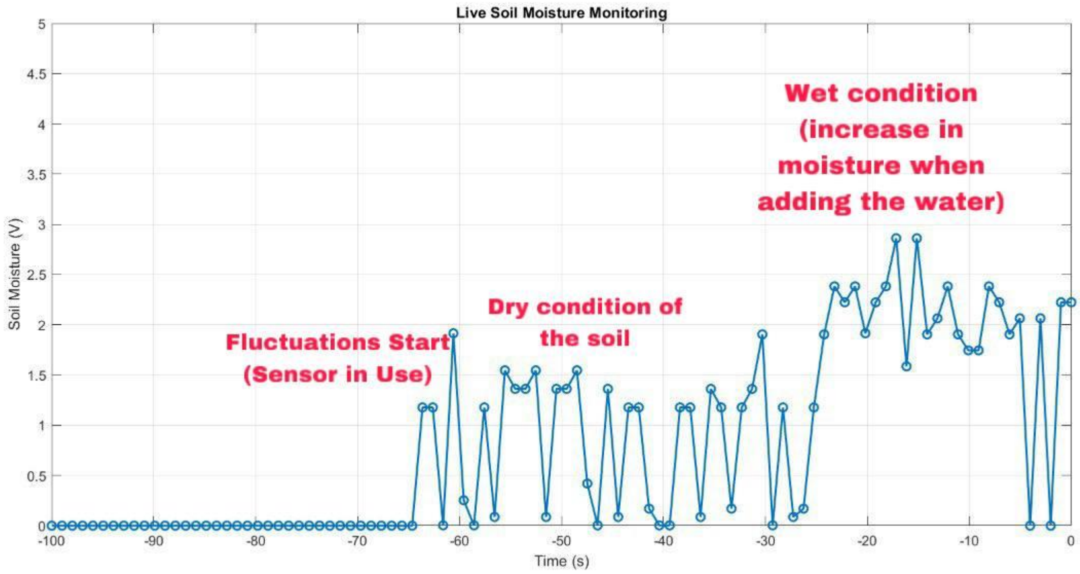
INTRODUCTION: This project involves developing an automated plant watering system using an Arduino-compatible Grove board and MATLAB to monitor and maintain soil moisture levels. **OBJECTIVES:** The goal is to ensure plants receive adequate water by measuring soil moisture and controlling a water pump based on real-time sensor data. **DESIGN ANALYSIS:** The system includes a soil moisture sensor, a MOSFET relay, and a water pump, integrated with a MATLAB program structured as a state machine to handle dry, wet, and saturated soil states. **PROCEDURE:** Soil moisture levels are monitored continuously via analog sensor readings, triggering the pump through digital signals when needed. The MATLAB program runs a live graph for visualization and ensures precise water delivery based on moisture thresholds. **RESULTS:** The implemented system successfully maintained desired soil moisture levels in a test environment, automating the watering process effectively. **CONCLUSIONS AND RECOMMENDATIONS:** The project demonstrates a simple, efficient automation technique for plant care. Further improvements, such as integrating wireless communication for remote monitoring, are recommended.

Plant and Hardware setup




Start Program
Read Soil Moisture Sensor Value
Is Soil Dry (Value < reallyDryValue)?
Yes
No
Turn On Pump
Is Soil Wet but Not Too Wet (Value < moistureThreshold)?
Yes
No
Turn On Pump
Is Soil Wet Enough (Value <= saturatedValue)?
Yes
Turn Off Pump
Wait (Pause)

Flowchart


Live Soil Moisture Monitoring
Soil Moisture (V)
Time (s)
Fluctuations Start
(Sensor in Use)
Dry condition of
the soil
Wet condition
(increase in
moisture when
adding the water)